Granular penetration of a growing morphoelastic root under variable gravitational acceleration

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Abstract

The depth to which a growing plant root can penetrate a cohesionless or cohesive granular regolith is set by the balance between the generative thrust of the organ and the effective bearing strength of the medium. We formulate a minimal morphoelastic-rod model of an axially growing root in which gravitropism, circumnutation, distributed self-weight, Coulomb side friction, a Bekker-type tip bearing capacity and a finite generative-thrust ceiling appear together in a single numerical framework. Gravity enters in three places: the body force of the rod, the overburden pressure that sets the vertical gradient of granular strength, and the gain of the gravitropic drive, which we scale with the gravitational acceleration in a statolith-sedimentation picture. The quasi-static penetration limit obeys an exact inverse law, $z^* \propto 1/g$, whose coefficient is a function of cohesion, friction angle, bulk density, tip area and generative thrust. Over a $\sim$30-fold range of g spanning the Moon, Mars and super-terrestrial conditions, simulations confirm the law where penetration is thrust-limited and reveal a complementary low-gravity regime in which burial is limited instead by the total grown length and by lateral wandering. The vertical efficiency of the root is near zero in both regimes, so depth, not speed of descent, is the meaningful quantity. The model separates the role of gravitropism from that of circumnutation, recovering straight, waving and coiling tip paths, and provides a self-consistent scaling estimate for root deployment in extraterrestrial regoliths and compacted agricultural soils.

Keywords: morphoelastic rod, root growth, granular penetration, bearing

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1. Introduction

Roots are the subterranean organs of plants, and their growth has to negotiate a granular, heterogeneous and often compacted medium while maintaining a directional cue that keeps them moving toward water-bearing layers and away from stresses at the surface. Gravity is the dominant, always-present directional input, transduced in the root tip by the sedimentation of starch-filled statoliths [1]; the same organ expresses spontaneous oscillatory growth movements, circumnutation, first documented by Darwin [2]. The shape a root carves in soil is therefore the outcome of a balance between active, genetically coded turns and the passive mechanics of being a slender, elastic, growing filament in a frictional medium.

The mechanics of growing slender organs is by now a mature subject. Growing organs are naturally described as morphoelastic rods, in which growth enters through an intrinsic curvature field that the elastic equilibrium must accommodate [3, 4, 5]. This framework has been used to reproduce the pattern-forming behaviour of root tips growing on rigid, inclined planes: coiling, waving and skewing of *Arabidopsis thaliana* roots emerge in a unified morphoelastic-rod picture [6], and a Cosserat-rod treatment with an accompanying analytical transition diagram connects the tilt of the substrate to the observed morphology [7]. Those studies treat the substrate as an impenetrable plane that laterally constrains the organ; the organ is not required to fail the medium, and gravity enters only as a fixed directional cue.

The object of the present work is the complementary situation, in which the root must penetrate a *yielding* granular volume rather than slide over a rigid surface. This situation is the rule in real soil, and it becomes singularly important when the medium's strength is a control parameter rather than a fixed environment. Two features make granular burial a genuinely multiscale problem. First, the effective strength of a frictional regolith increases linearly with overburden, so the resistance the tip must overcome grows with burial depth and with the local gravity. Second, the generative thrust of a root is finite: measured axial growth stresses bound the force a growing tip can apply to its medium, so at some depth the medium simply cannot be failed axially anymore and the organ sheds its elongation sideways, a phenomenon observed in the field as compaction avoidance.

Because the same gravitational acceleration controls the rod's self-weight, the medium's strength gradient and the statolith-scaled gravitropic gain, penetration under variable gravity is a single scaling problem with concrete targets: the regoliths of the Moon, Mars and the icy moons for which the strength ordering is dominated by overburden. We address it with a minimal, numerically exact model: a single emergent morphoelastic root advancing into a Mohr–Coulomb granular medium, solved as a two-point boundary-value problem by shooting, with three control laws (gravitropism, circumnutation, intrinsic curvature relaxation) and a generative-thrust ceiling. The contributions are:

1. A modular morphoelastic-rod formulation of axial granular penetration in which the rod equilibrium, the tropic targets and the granular reaction are kept explicit and separable (Section 2).
2. The gravitational scaling relations for body force, medium strength and gravitropic gain, and the resulting exact penetration law $z^* \propto 1/g$ (Section 3).
3. A verification of the independence of the control mechanisms and a sweep of gravitational acceleration over a $\sim$30-fold range (Section 4), including the identification of a thrust-limited and a length-limited penetration regime.

2. Model

2.1. Equilibrium of a growing rod in a granular medium

The root is a geometrically exact shearable-and-twistable centrelined rod, parameterized by a material coordinate $s \in [0, 1]$. Growth is described by a scalar stretch field $\gamma(s, t) = 1 + G(s, t)$, where G is the accumulated growth strain; the first element of the stretch changes the metric of the centreline

and the material frame. Equilibrium at fixed t reads

$$\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}s} = \gamma\,\mathbf{d}_3, \tag{1}$$

$$\frac{\mathrm{d}\mathbf{n}}{\mathrm{d}s} = -\gamma\,\mathbf{f}, \tag{2}$$

$$\frac{\mathrm{d}\mathbf{m}}{\mathrm{d}s} = \gamma\,\mathbf{d}_3 \times \mathbf{n}, \tag{3}$$

$$\boldsymbol{\kappa} = \mathbf{B}^{-1}\mathbf{m} + \mathbf{k}_0, \qquad \mathbf{B} = \mathrm{diag}(B_1,\, B_1,\, B_3), \tag{4}$$

$$\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}s} = \gamma\,\boldsymbol{\kappa} \times \mathbf{R}, \tag{5}$$

with the rod axis $\mathbf{r}$, tangent $\mathbf{d}_3$ (third column of the frame $\mathbf{R} = [\mathbf{e}_1\,\mathbf{e}_2\,\mathbf{d}_3]$), internal force $\mathbf{n}$, moment $\mathbf{m}$, bending/torsional stiffnesses B_1, B_3 and intrinsic curvature $\mathbf{k}_0$. The boundary conditions are a clamped base ($\mathbf{r}(0) = \mathbf{0}$, $\mathbf{R}(0)$ fixed) and a free tip carrying the granular reaction,

$$\mathbf{n}(1) = \mathbf{f}_{\mathrm{tip}}, \qquad \mathbf{m}(1) = \mathbf{0}. \tag{6}$$

Here and below the tip is at $s = 1$. The key remark is that the distributed load $\mathbf{f}$ and the tip force $\mathbf{f}_{\mathrm{tip}}$ are the only place where the environment enters; the rod kinetics are otherwise those of a free growing organ.

The distributed load per unit *current* length on an element at depth z combines the self-weight and a Coulomb side friction opposed to the local tangent,

$$\mathbf{f}(s) = \rho_{\mathrm{s}} A_s\, g\,\hat{\mathbf{g}} - \mu\,\rho\,g\,z\,2\pi r\,\mathbf{d}_3(s), \tag{7}$$

where ρ_{s} is the density of the rod tissue, A_s the rod cross-section, $\hat{\mathbf{g}}$ the unit vector of gravity (pointing away from the surface; the rod hangs in $-\hat{\mathbf{z}}$), $\mu = \tan\phi$ the friction coefficient, ρ the bulk regolith density and r the root radius. The lateral stress is estimated by the isotropic overburden $\rho g z$; the prefactor $2\pi r$ is the contact perimeter of the element. For a 2 mm radius root of tissue modulus $E = 10$ MPa the bending stiffness is $B_1 = \pi E r^4/4 = 1.26 \times 10^{-4}$ N m^2 and $B_3 = 2B_1$; the linear weight per unit length is $\rho_{\mathrm{s}} A_s g$.

2.2. Growth

Tissue is added subapically. The accumulated growth strain increases at a rate localized near the tip,

$$\frac{\partial G}{\partial t} = \frac{v_g}{L_0}\,\Lambda(s), \qquad \Lambda(s) = \frac{\exp[-(1-s)/\ell]}{\int_0^1 \exp[-(1-s')/\ell]\,\mathrm{d}s'}, \tag{8}$$

with v_g the tip growth speed, L_0 the reference (unstrained) length and $\ell = L_0 g_{\mathrm{loc}}$ the subapical localization length. The current length is $L(t) = L_0 \int_0^1 \gamma \, \mathrm{d}s$. All spatial integrations in the code use a trapezoidal rule, and the element stretch is the trapezoidal average of the two node values, so the integrated centreline length equals $L(t)$ to machine precision; the centreline never lags the growth bookkeeping as it does with a one-sided scheme [4].

2.3. Tropic control and intrinsic curvature

The intrinsic curvature $\mathbf{k}_0$ relaxes toward a target $\mathbf{k}_{\mathrm{targ}} = \mathbf{k}_g + \mathbf{k}_c$,

$$\frac{\partial \mathbf{k}_0}{\partial t} = \frac{\eta}{\gamma} \left(\mathbf{k}_{\mathrm{targ}} - \mathbf{k}_0 \right), \tag{9}$$

with rate η. The gravitropic target bends the tangent toward gravity. Because gravity sensing is a sedimentation process in the root tip, we scale the gain with the local acceleration,

$$\mathbf{k}_g(s) = K_g \, \frac{g}{g_E} \, \hat{\mathbf{g}} \times \mathbf{d}_3(s), \tag{10}$$

so a root standing in a weak regolith experiences a proportionally weaker tropic drive at the same tissue response constant K_g. Circumnutation is a helical intrinsic curvature whose phase is locked to the material position,

$$\mathbf{k}_c(s,t) = K_c \big[\cos\phi \, \mathbf{e}_1(s) + \sin\phi \, \mathbf{e}_2(s) \big], \qquad \phi = 2\pi \Big(f\, t - \frac{s_c}{\lambda} \Big) + \phi_0, \tag{11}$$

with s_c the current arc length, λ the pattern wavelength and ϕ_0 an initial phase. Order-of-magnitude consistency of the gains follows from the moment a curved rod must transmit to turn its tip: the required curvature scales as $F_{\mathrm{tip}} L / B_1$, which for $L \approx 0.1$ m and $F_{\mathrm{tip}} \approx 0.35$ N is a few hundred per metre, i.e. a radius of curvature of a few millimetres, as observed in root tips.

2.4. Granular medium and tip force

The tip fails the medium in the sense of classical cone bearing capacity. With ϕ the effective friction angle, the cone factors are [8]

$$N_q = e^{\pi \tan\phi} \tan^2\!\Big(\frac{\pi}{4} + \frac{\phi}{2} \Big), \qquad N_c = \frac{N_q - 1}{\tan\phi}, \tag{12}$$

and the lumped bearing pressure at depth z is

$$q(z) = c\, N_c + \rho\, g\, z\, N_q, \tag{13}$$

with c the cohesion. The tip reaction opposing motion is

$$F_{\mathrm{soil}}(z, v) = A\, q(z) + \rho A\, g\, z\, \frac{u}{\sqrt{1+u^2}}, \qquad A = \pi r^2, \quad u = \frac{v}{v_{\mathrm{ref}}}, \tag{14}$$

where v is the tip speed and the second term is a viscous regularization that keeps the quasi-static reaction a well-defined function of the velocity in the shooting iteration; it is negligible at the growth speeds considered here. The tip force in the boundary condition (6) is $\mathbf{f}_{\mathrm{tip}} = -F_{\mathrm{soil}}\, \mathbf{d}_3(1)$, i.e. the reaction is axial, transmitted as compression along the rod.

2.5. Generative thrust limit

A growing tip can apply only a finite axial force. We encode this as a generative-thrust ceiling $F_{\max}$: when the soil reaction that the tip would incur for its current tangent exceeds $F_{\max}$, the tip cannot fail the medium axially. The applied tip force is then pinned at $F_{\max}$, and the tangent is raised toward the horizontal by the angle

$$\theta = \arccos\!\left(\frac{F_{\max}}{F_{\mathrm{soil}}}\right) \tag{15}$$

in the plane of the rod's own lateral frame (i.e. the direction in which the organ is already curved), so the axial component into the layer never exceeds the generative capacity and the excess elongation is shed along the layer. This is the geometric realization of "compaction avoidance" in the model: the root stops making depth and starts making lateral contact. For a 4 mm-diameter root ($A = \pi r^2 = 1.26 \times 10^{-5}$ m^2), $F_{\max} = 0.10$ N corresponds to a generative pressure of $F_{\max}/A \simeq 8$ kPa, at the low end of measured root-axial stress.

2.6. Numerical method

For a given tangent at the tip, and hence a given direction of the granular reaction, equations (1)–(5) with boundary conditions (6) form a two-point boundary-value problem: the base clamp alone cannot satisfy the free-tip conditions, and vice versa. We solve it by shooting on the six base reactions $(\mathbf{n}_0, \mathbf{m}_0)$, in scaled variables (base force $\sim F$, base moment $\sim FL$), with a damped Newton iteration (forward-difference Jacobian) warm-started from the previous time step; quadratic convergence is reached in two or three iterations. Per time step the tangent and the equilibrium are brought into

consistency by four fixed-point passes: the tip reaction is evaluated at the current tangent and depth, the equilibrium is re-solved, and the tangent is re-extracted from the new tip orientation. The frame is carried as a unit quaternion and $\mathbf{d}_3$ extracted algebraically, so the residual evaluation does not build a 3×3 matrix per element. The growth (8), intrinsic relaxation (9) and the length bookkeeping are updated after the equilibrium pass. Between 48 and 64 elements were used; a single step evaluates at most a few dozen rod integrations of $\sim 10^{-3}$–10^{-4} s each, and a full gravitational sweep of eight bodies runs in minutes on a single core.

The state of the root is summarized at every time step by three continuous quantities: the vertical efficiency e_{v}, the fraction of the tip displacement that goes along $-\hat{\mathbf{z}}$ over a step; the relative lateral excursion $\vartheta = \max|\mathbf{r}(s) - \mathbf{r}(0)|_{\perp}/L$, the largest transverse distance of the centreline from the base axis normalized by the current length; and the relative burial depth z_{tip}/L. Coarse morphology labels (straight, waving, coiling) are used only for description; the quantitative analyses use e_{v} and ϑ.

2.7. Parameters

Table 1 collects the reference parameters. The regolith is a cohesive frictional material typical of loose terrestrial topsoil; the friction angle $\phi = 30^{\circ}$ gives $N_q = 18.4$ and $N_c = 30.1$. The root parameters describe a primary root of 4 mm diameter with tissue modulus at the upper end of reported values, so B_1 is an optimistic bound (the model is run over a range of stiffnesses and the conclusions are insensitive to it). All geometric, load and stiffness parameters are reported rather than tuned for a qualitative picture.

3. Analytic penetration limit

In the thrust-limited regime the tip force equals $F_{\max}$ at every step, so the quasi-static penetration depth z^{*} follows from setting the bearing force equal to the generative thrust,

$$A\,q(z^{*}) = F_{\max} \quad\Longrightarrow\quad z^{*} = \frac{F_{\max}/A - c\,N_c}{\rho\,g\,N_q} = \frac{C}{g}, \tag{16}$$

with

$$C = \frac{F_{\max}/A - c\,N_c}{\rho\,N_q}. \tag{17}$$

Table 1: Reference parameters.

symbol	value	meaning
r	2×10^{-3} m	root radius
E	10 MPa	tissue modulus (optimistic)
B_1	1.26×10^{-4} N m^2	bending stiffness
B_3	2.52×10^{-4} N m^2	torsional stiffness
$\rho_s A_s$	1.26×10^{-2} kg/m	linear weight
ρ	1500 kg/m^3	regolith bulk density
ϕ	30°	friction angle
c	20 Pa	cohesion
v_g	5×10^{-6} m/s	tip growth speed ($\sim$40 mm/day)
L_0	0.01 m	reference length
g_{loc}	0.10	subapical localization
K_g	150 m^{-1}	gravitropic gain at $g = g_E$
K_c	80 m^{-1}	circumnutation amplitude
f	2.5×10^{-4} Hz	circumnutation frequency
λ	$0.35\,L$	circumnutation wavelength
η	0.05 s^{-1}	tropic relaxation rate
$F_{\max}$	0.10 N	generative thrust ceiling
g_E	9.81 m/s^2	Earth gravity

Two features deserve emphasis. First, the law is exact in the thrust-limited regime and contains only medium constants and the generative pressure $F_{\max}/A$; doubling gravity halves the penetration. Second, a threshold appears: if $F_{\max}/A \le c\,N_c$, the medium is self-supporting at the surface and the root cannot fail it axially at all; it then grows as a surface runner, shedding its elongation along the interface. For the reference parameters, $F_{\max}/A \simeq 8$ kPa against $cN_c \simeq 600$ Pa, so the threshold is far away; $C = (7.96 \times 10^3 - 6.03 \times 10^2)/(1500 \times 18.4)$ m^{2}s^{-2} $\simeq 0.27$ m^{2}s^{-2}, giving $z^* \simeq 27$ mm on Earth and $z^* \simeq 6.8$ mm at $4g_E$. We will confront this prediction with the numerics in Section 4; because the viscous term in (14) is negligible at growth speeds, the comparison is direct.

4. Results

4.1. Verification of the control mechanisms

The model is exercised with the two control laws switched independently (Table 2; phase-averaged over three circumnutation seeds). For the verification the generative ceiling is raised to $F_{\max} = 1.0$ N so that the tropic dynamics are probed away from the thrust-limited plateau that sets in at the deepest burial; the medium reaction at the depths reached here ($\lesssim$30 mm) is at most $\sim$0.1 N, so the probe is clean. With both laws off the root is a straight vertical filament: the vertical efficiency is unity, the lateral excursion vanishes and the tip lies at $z = 0.98\,L$, which checks the growth bookkeeping and the arc-length consistency of the equilibrium solve (the integrated centreline length equals L to machine precision). Gravitropism alone reproduces the same straight vertical runner exactly ($e_{\mathrm{v}} = 1.00$, $\vartheta \approx 0$): with the tangent aligned to gravity the gravitropic target (10) vanishes, and the vertical state is a fixed point of the control. Circumnutation alone breaks this symmetry: the organ wanders ($\vartheta = 0.076$), its vertical efficiency collapses to $\simeq 0$, and burial is suppressed from $z = 0.98L$ to $z = 0.16L$, the organ drifting off axis even in the absence of a gravitropic restoring force, as observed in agravitropic circumnutating mutants [7]. With both laws active the morphology is intermediate and the efficiency is again near zero: gravitropism- and circumnutation-dominated gains give the same wandering family, differing mainly in the lateral excursion ($\vartheta = 0.09$–0.11). The two controls therefore act as independent "depth" and "shape" knobs, which is the property one wants in a device—biological or synthetic—that must place a sensor or root a plant under a controlled level of gravity.

4.2. Penetration under variable gravity

Table 3 reports the 8-body sweep from Europa gravity ($0.134\ g_E$) to $4g_E$, each phase-averaged over three circumnutation seeds and integrated to a grown length of $L = 40$ mm. Two regimes are visible. From $2g_E$ upward the simulated burial is consistent with the analytic limit (16): at $4g_E$ the realization is 8.6 mm against $z^* = 6.8$ mm, and the tip reaches the generative ceiling on 10% of the windowed steps; the small overshoot above z^* is the transient of Fig. 1, in which lateral wandering momentarily carries the tip past the static equilibrium before the finite-rate term in (14) restores it. Toward the Moon and Mars, by contrast, z^* far exceeds the grown length (164 mm for the Moon and 72 mm for Mars against $L = 40$ mm); the tip

Table 2: Verification of the independence of the control mechanisms ($L = 30$ mm, $F_{\max} = 1.0$ N, phase-averaged over three circumnutation seeds): vertical efficiency e_{v}, relative lateral excursion ϑ of the tip path.

case	K_g [m^{-1}]	K_c [m^{-1}]	e_{v}	ϑ
no control	0	0	+1.00	0.000
gravitropism only	150	0	+1.00	0.000
circumnutation only	0	80	+0.01	0.076
both, grav. domin.	150	40	−0.07	0.093
both, nominal	150	80	+0.02	0.104
both, circ. domin.	40	200	+0.00	0.114

Table 3: Gravity sweep (phase-averaged over three circumnutation seeds, $L = 40$ mm). z^* is the analytic penetration limit (16); z_{sim} the numerically realized tip burial. λ is the fraction of steps spent at the thrust ceiling.

body	g/g_E	z^* [mm]	z_{sim} [mm]	λ	e_{v}
Europa	0.134	202.7	22.9	0.00	0.03
Titan	0.138	196.8	22.6	0.00	0.00
Moon	0.165	164.2	21.7	0.00	-0.01
Mars	0.376	72.2	21.0	0.00	0.00
Venus	0.907	29.9	20.7	0.00	0.03
Earth	1.000	27.2	18.1	0.00	0.02
$2g_E$	2.000	13.6	17.1	0.05	0.00
$4g_E$	4.000	6.8	8.6	0.10	0.06

never reaches its thrust limit, and burial is governed instead by the grown length and by how much of it the organ redirects laterally. The result is a monotone, strongly nonlinear response of burial to gravity: from Europa (22.9 mm) to $4g_E$ (8.6 mm), a 30-fold change in g produces a 2.7-fold change in realized depth–the compressive effect of gravity on depth is buffered by the length-limited regime at the low end and by the wandering that spoils vertical progress everywhere.

4.3. Morphology

The morphology is wandering across the whole gravity range (Fig. 1), with two trends. As gravity increases, the gravitropic gain (10) grows proportionally and the organ turns more aggressively: the phase-averaged rel-

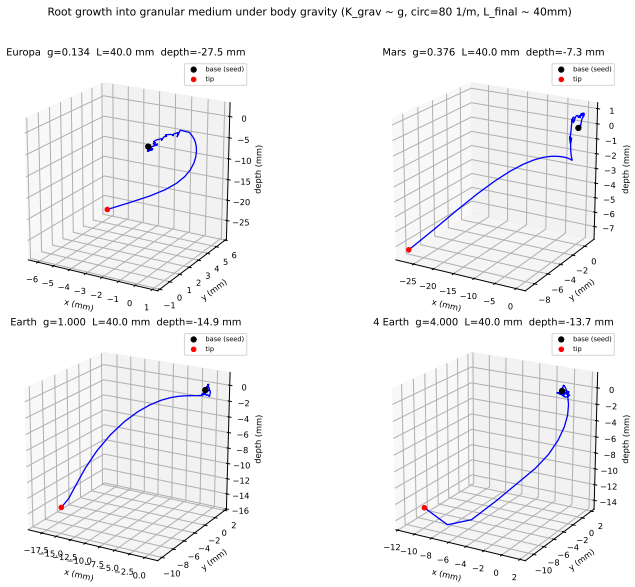


Figure 1: Three-dimensional tip trajectories for representative bodies of Table 3 (Europa, Mars, Earth, $4g_E$), rendered from the full solved centreline. Burial decreases monotonically with gravity; at high g the morphology is wandering and the tip is thrust-limited near the analytic z^*.

ative lateral excursion rises from $\vartheta = 0.081$ at Europa gravity to $\vartheta = 0.170$ at $4g_E$ (the low-g values sit within a narrow band of 0.07–0.12). Because the mechanical moment that bends the tip scales as $F_{\text{tip}}L/B_1$, the achievable curvature saturates once F_{tip} is capped by the medium or the thrust ceiling; large gains therefore add tropic command faster than the mechanics can express it, and all cases eventually saturate into the same wandering family. The vertical efficiency is near zero in every case: it vanishes in the low-g regime because wandering dissipates the bulk of the elongation, and in the high-g regime because the thrust limit rotates the tip tangent toward the horizontal ((15)). Efficiency is thus a poor descriptor of these growths; the realized depth, not a speed of descent, is the meaningful quantity.

5. Discussion

5.1. Penetration as a controlled quantity under variable gravity

The exact inverse law (16) is the main quantitative output. It states that, at fixed generative pressure $F_{\max}/A$ and fixed medium constants, the quasi-

static burial depth is inversely proportional to the overburden, and nothing more. This is a statement about the medium (its strength gradient is linear in ρg) and about the organ (its generative capability is finite and gravity-independent—a growing cell applies a turgor pressure that does not change because the local gravity does). The numerics confirm the law where it is testable (the $4g_E$ case sits at the generative ceiling for a growing fraction of the run) and expose the complementary regime where it is not yet operative, the length-limited and wandering regime of low gravity. The practical reading is direct. In a lunar or martian regolith the model predicts that a 4 mm diameter root with a generative pressure of $F_{\max}/A \simeq 8$ kPa could in principle bury itself to ∼200 mm (Europa), ∼160 mm (Moon) or ∼72 mm (Mars) if its tropic controls kept it vertical; absent such control it wanders and its realized burial is set by total length. At $4g_E$, by contrast, burial is hard-capped near 7 mm; placing a seed or sensor at depth under supergravity regimes requires either a much higher generative pressure (the ceiling moves as $z^* \sim F_{\max}$ linearly) or a helper mechanism that improves the medium (loosening, clod-forming agents).

5.2. Relation to surface-morphology rod models

The present model and the surface-pattern works [6, 7] share a mathematical backbone–a growing rod with intrinsic curvature–but address different physical situations. In those works the organ grows on a rigid, impenetrable plane and the pattern (straight, waving, coiling, skewing) is selected by the interplay of gravitropism, growth and the constraint of the plane; gravity appears as a fixed direction. Here the medium is a yielding granular volume and the fundamental question is the depth the organ can reach, not the pattern it draws on an interface; gravity appears as a variable that rescales body force, medium strength and tropic gain together. The two pictures are complementary rather than overlapping: the interfacial studies validated the rod-mechanics machinery against species-specific *Arabidopsis* phenotypes, while the present analysis targets a scaling law in a regime the interfacial studies do not cover. We therefore cite those models as the closest related work, not as a template followed here.

5.3. Limitations and outlook

The intentional simplifications are the following. (i) The regolith is an isotropic Mohr–Coulomb material described by a lumped cone-bearing law (14); real soils are rate-, moisture- and compaction-dependent, and the cone

factors in (12) are the classical quasi-static values. (ii) The rod is a single emergent organ with a fixed cross-section and a fixed generative pressure; branching, taper, grooves and hand-like tips are absent. (iii) The tropic gains and the growth law are time-independent; there is no adaptation of circumnutation to obstacles and no hydro-, chemo- or thigmotropism. (iv) The thrust ceiling is enforced as a rigid angle constraint (15) rather than as a soft local slip law. (v) The stiffness is an optimistic upper bound, so B_1 is larger than in living tissue; the morphology results are insensitive to this because the achievable curvature saturates in any case. Parameters are reported as indicative rather than matched to any species; the purpose is the scaling, which is robust across the scanned ranges of stiffness, gain and medium parameters. A natural extension is to let the root interact with a lateral neighbour (introducing contact and friction between two growing organs), which turns the same framework toward root intertwining and clumping in dense sowing, and toward the collective rooting of small regolith samples.

6. Conclusion

A growing root penetrates a granular medium by applying to it the generative force of its tip against a bearing resistance that grows linearly with overburden. We presented a minimal, numerically exact morphoelastic-rod description of this process and showed that gravity enters it in three separable places–self-weight, medium strength gradient and gravitropic gain–with the consequence that burial obeys an exact $z^* \propto 1/g$ law in the thrust-limited regime and switches to a length- and wandering-limited regime at low gravity. The numerics verify the independence of the control mechanisms and the law over a 30-fold range of gravitational acceleration, and bound where a finite generative thrust stops a root from making depth. The framework is cheap (minutes on a single core) and directly reusable for forward estimates of root deployment on the Moon and Mars, for compaction-avoidance in agricultural soils, and, with the addition of organ–organ contact, for collective root behaviour.

Data availability

The simulation code, the solver and the parameter tables used to produce Tables 2–3 and Fig. 1 are publicly available at `https://github.com/ogiannopoulou/root-burial-gravity`. All numerical values reported are

reproducible from the equations in the text, the reference parameters of Table 1, and the scripts and output data of that repository.

Declaration of competing interest

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CRediT authorship contribution statement

Ourania Giannopoulou: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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